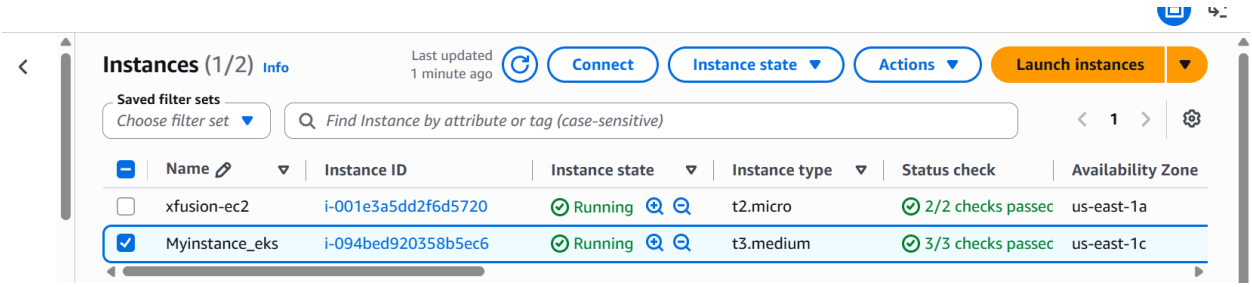
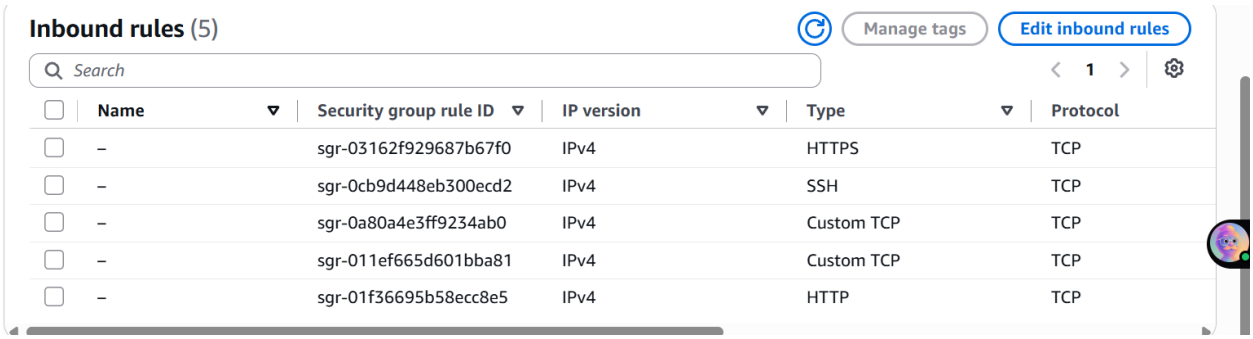
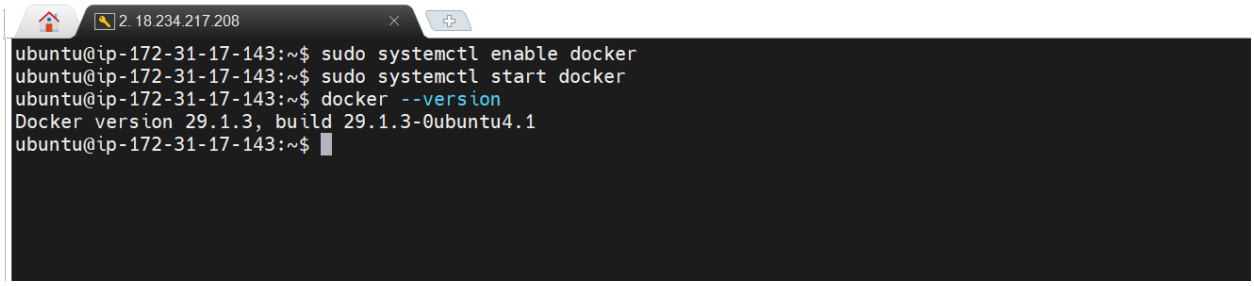
Step 1:

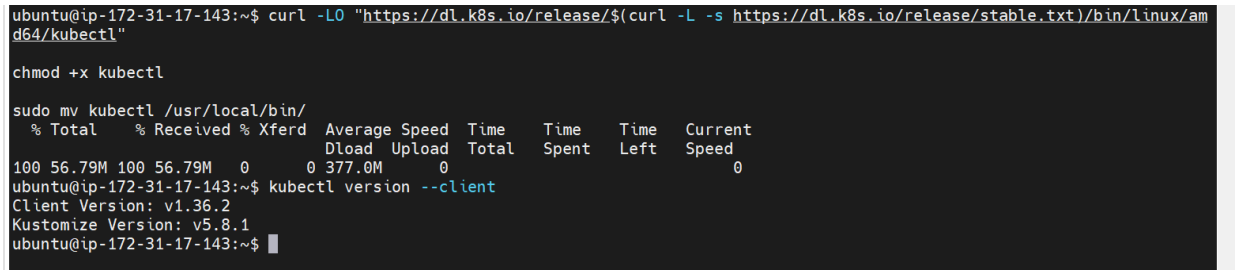




Step 2:



Step 3:



```
ubuntu@ip-172-31-17-143:~$ curl -LO https://storage.googleapis.com/minikube/releases/latest/minikube-linux-amd64
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100 128.6M 100 128.6M    0     0 244.6M      0                              0
ubuntu@ip-172-31-17-143:~$ sudo install minikube-linux-amd64 /usr/local/bin/minikub
ubuntu@ip-172-31-17-143:~$ minikube version
minikube: command not found
ubuntu@ip-172-31-17-143:~$ sudo install minikube-linux-amd64 /usr/local/bin/minikube
ubuntu@ip-172-31-17-143:~$ minikube version
minikube version: v1.38.1
commit: c93a4cb9311efc66b90d33ea03f75f2c4120e9b0
ubuntu@ip-172-31-17-143:~$
```

Starting the cluster

```
3.18.234.217.208
ubuntu@ip-172-31-17-143:~$ minikube start --driver=docker
* minikube v1.38.1 on Ubuntu 26.04
* Using the docker driver based on user configuration
! Starting v1.39.0, minikube will default to "containerd" container runtime. See #21973 for more info.

X The requested memory allocation of 3072MiB does not leave room for system overhead (total system memory: 3831MiB). You may
ace stability issues.
* Suggestion: Start minikube with less memory allocated: 'minikube start --memory=3072mb'

* Using Docker driver with root privileges
* Starting "minikube" primary control-plane node in "minikube" cluster
* Pulling base image v0.0.50 ...
* Downloading Kubernetes v1.35.1 preload ...
    > preloaded-images-k8s-v18-v1...:  272.45 MiB / 272.45 MiB  100.00% 285.35
    > gcr.io/k8s-minikube/kicbase...:  519.58 MiB / 519.58 MiB  100.00% 122.84
* Creating docker container (CPUs=2, Memory=3072MB) ...
* Preparing Kubernetes v1.35.1 on Docker 29.2.1 ...
* Configuring bridge CNI (Container Networking Interface) ...
* Verifying Kubernetes components...
  - Using image gcr.io/k8s-minikube/storage-provisioner:v5
* Enabled addons: storage-provisioner, default-storageclass
* Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
ubuntu@ip-172-31-17-143:~$
```

```
ubuntu@ip-172-31-17-143:~$ kubectl get nodes
NAME       STATUS   ROLES           AGE   VERSION
minikube   Ready    control-plane   26s   v1.35.1
ubuntu@ip-172-31-17-143:~$
```

```
ubuntu@ip-172-31-17-143:~$ minikube addons enable metrics-server
* metrics-server is an addon maintained by Kubernetes. For any concerns contact minikube on GitHub.
You can view the list of minikube maintainers at: https://github.com/kubernetes/minikube/blob/master/OWNERS
  - Using image registry.k8s.io/metrics-server/metrics-server:v0.8.1
* The 'metrics-server' addon is enabled
ubuntu@ip-172-31-17-143:~$
```

Create Namespace food app

```
ubuntu@ip-172-31-17-143:~$ kubectl create namespace food-app
namespace/food-app created
ubuntu@ip-172-31-17-143:~$ kubectl get ns
NAME              STATUS   AGE
default           Active   2m5s
food-app          Active   45s
kube-node-lease   Active   2m5s
kube-public       Active   2m5s
kube-system       Active   2m5s
ubuntu@ip-172-31-17-143:~$
```

Namespace.yaml

```
ubuntu@ip-172-31-17-143:~/food-ordering-k8s/yaml$ cat namespace.yaml
apiVersion: v1
kind: Namespace

metadata:
  name: food-app
ubuntu@ip-172-31-17-143:~/food-ordering-k8s/yaml$
```

```
ubuntu@ip-172-31-17-143:~/food-ordering-k8s/yaml$ kubectl get namespaces
NAME              STATUS   AGE
default           Active   10m
food-app          Active   9m4s
kube-node-lease   Active   10m
kube-public       Active   10m
kube-system       Active   10m
ubuntu@ip-172-31-17-143:~/food-ordering-k8s/yaml$
```

Cnfigmap.yaml

```
ubuntu@ip-172-31-17-143:~/food-ordering-k8s/yaml$ cat configmap.yaml
apiVersion: v1
kind: ConfigMap

metadata:
  name: app-config
  namespace: food-app

data:
  DB_HOST: mysql-service
  DB_PORT: "3306"
  APP_ENV: production
ubuntu@ip-172-31-17-143:~/food-ordering-k8s/yaml$
```

```
ubuntu@ip-172-31-17-143:~/food-ordering-k8s/yaml$ kubectl apply -f configmap.yaml
configmap/app-config created
```

```
3. 18.234.217.208
ubuntu@ip-172-31-17-143:~/food-ordering-k8s/yaml$ kubectl get configmap -n food-app
NAME               DATA   AGE
app-config         3      27s
kube-root-ca.crt   1      11m
ubuntu@ip-172-31-17-143:~/food-ordering-k8s/yaml$ kubectl describe configmap app-config -n food-app
Name:         app-config
Namespace:    food-app
Labels:       <none>
Annotations:  <none>

Data
====
APP_ENV:
----
production

DB_HOST:
----
mysql-service

DB_PORT:
----
3306


BinaryData
====

Events:  <none>
```

Secret.yaml

```
3. 18.234.217.208
ubuntu@ip-172-31-17-143:~/food-ordering-k8s/yaml$ touch secrets.yaml
ubuntu@ip-172-31-17-143:~/food-ordering-k8s/yaml$ nano secrets.yaml
ubuntu@ip-172-31-17-143:~/food-ordering-k8s/yaml$ cat secrets.yaml
apiVersion: v1
kind: Secret
metadata:
  name: mysql-secret
  namespace: food-app

type: Opaque

stringData:
  MYSQL_ROOT_PASSWORD: admin123
  MYSQL_DATABASE: fooddb
ubuntu@ip-172-31-17-143:~/food-ordering-k8s/yaml$
```

```
ubuntu@ip-172-31-17-143:~/food-ordering-k8s/yaml$ kubectl apply -f secrets.yaml
secret/mysql-secret created
ubuntu@ip-172-31-17-143:~/food-ordering-k8s/yaml$
```

My sql Deployment.yaml

```
ubuntu@ip-172-31-17-143:~/food-ordering-k8s/yaml$ cat mysql-deployment.yaml
apiVersion: apps/v1
kind: Deployment

metadata:
  name: mysql
  namespace: food-app

spec:
  replicas: 1

  selector:
    matchLabels:
      app: mysql

  template:

    metadata:
      labels:
        app: mysql

    spec:

      containers:

      - name: mysql

        image: mysql:8.0

        ports:

        - containerPort: 3306
```

3. 18.234.217.208

```
  template:

    metadata:
      labels:
        app: mysql

    spec:

      containers:

      - name: mysql

        image: mysql:8.0

        ports:

        - containerPort: 3306

        env:

        - name: MYSQL_ROOT_PASSWORD
          valueFrom:
            secretKeyRef:
              name: mysql-secret
              key: MYSQL_ROOT_PASSWORD

        - name: MYSQL_DATABASE
          valueFrom:
            secretKeyRef:
              name: mysql-secret
              key: MYSQL_DATABASE
```

my-service.yaml

```
ubuntu@ip-172-31-17-143:~/food-ordering-k8s/yaml$ cat mysql-service.yaml
apiVersion: v1
kind: Service

metadata:
  name: mysql-service
  namespace: food-app

spec:

  selector:
    app: mysql

  ports:

  - port: 3306

    targetPort: 3306

  type: ClusterIP
ubuntu@ip-172-31-17-143:~/food-ordering-k8s/yaml$
```

app-deployment.yaml

```
ubuntu@ip-172-31-17-143:~/food-ordering-k8s/yaml$ cat app-deployment.yaml
apiVersion: apps/v1
kind: Deployment

metadata:
  name: food-app
  namespace: food-app

spec:

  replicas: 2

  selector:

    matchLabels:

      app: food-app

  template:

    metadata:

      labels:

        app: food-app

    spec:

      containers:
```

```
    labels:

      app: food-app

  spec:

    containers:

    - name: food-app

      image: madhu934652/food-app:latest

      ports:

      - containerPort: 8080

      envFrom:

      - configMapRef:
          name: app-config

      resources:

        requests:
          cpu: "200m"
          memory: "256Mi"

        limits:
          cpu: "500m"
          memory: "512Mi"

ubuntu@ip-172-31-17-143:~/food-ordering-k8s/yaml$
```

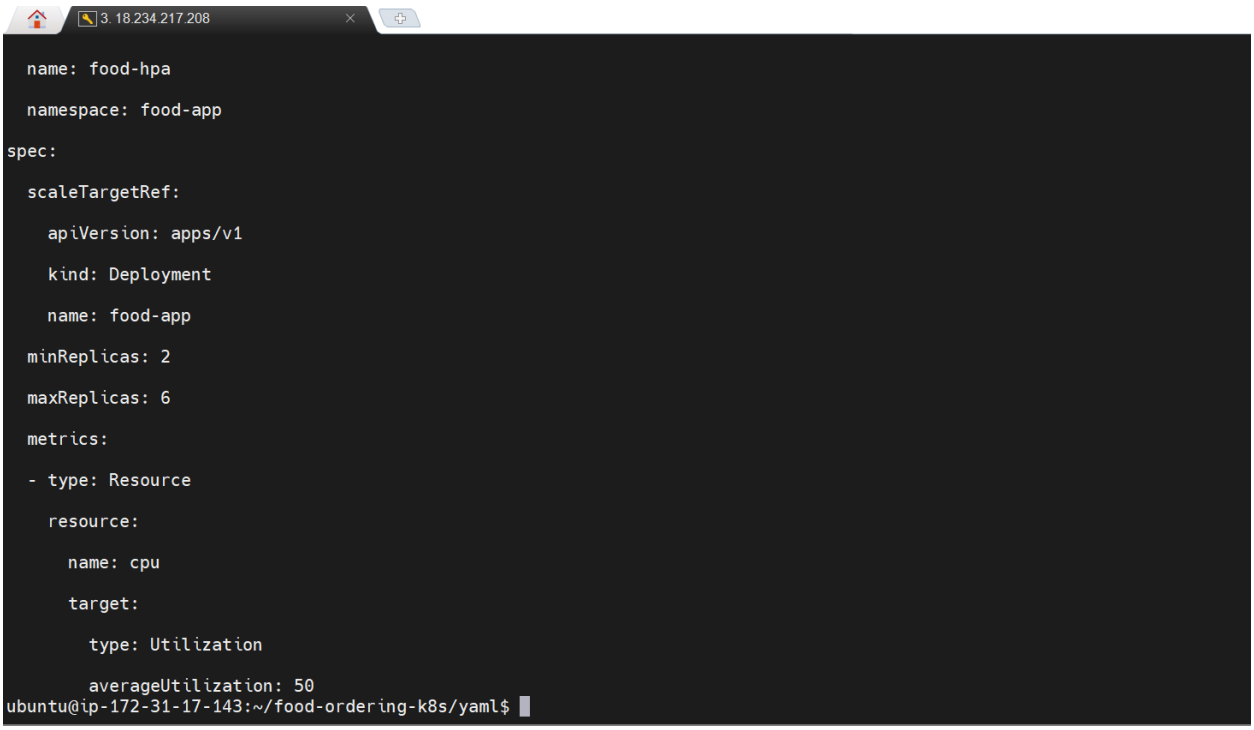
app-service.yaml

```
ubuntu@ip-172-31-17-143:~/food-ordering-k8s/yaml$ cat app-service.yaml
apiVersion: v1
kind: Service

metadata:

  name: food-service

  namespace: food-app

spec:

  selector:

    app: food-app

  ports:

  - port: 80

    targetPort: 8080

  type: NodePort
ubuntu@ip-172-31-17-143:~/food-ordering-k8s/yaml$
```

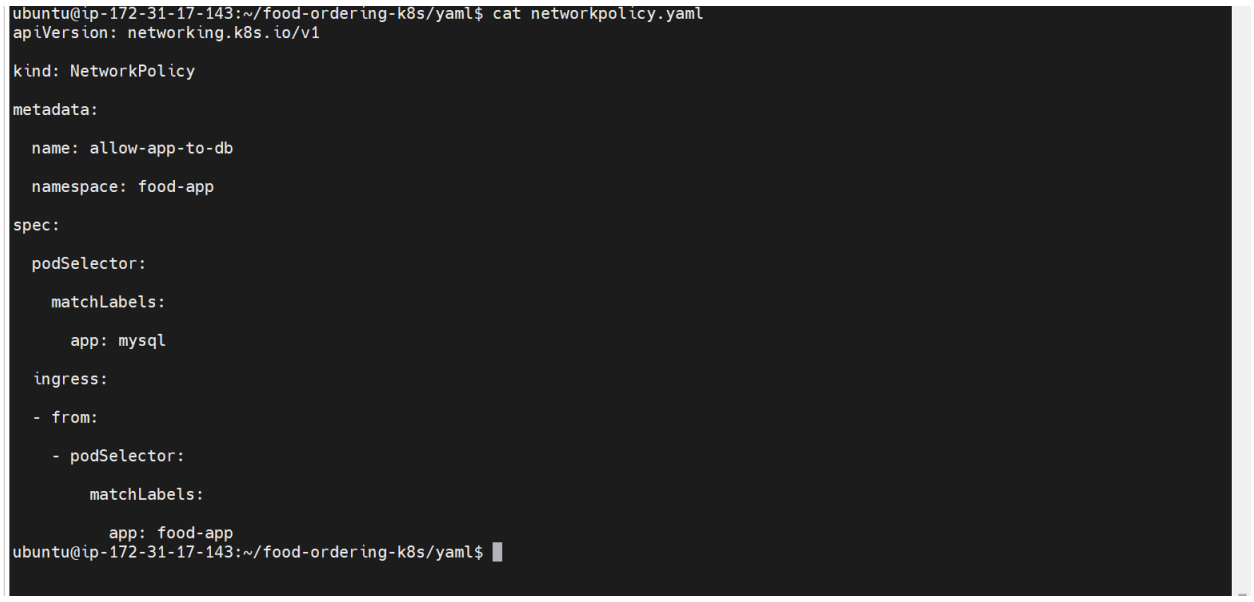
```yaml
kind: Ingress
metadata:
  name: food-ingress
  namespace: food-app
spec:
  rules:
  - host: food.local
    http:
      paths:
      - path: /
        pathType: Prefix
        backend:
          service:
            name: food-service
            port:
              number: 80
```

Hpa.yaml

```yaml
kind: HorizontalPodAutoscaler
metadata:
  name: food-hpa
  namespace: food-app
spec:
  scaleTargetRef:
    apiVersion: apps/v1
    kind: Deployment
    name: food-app
  minReplicas: 2
  maxReplicas: 6
  metrics:
  - type: Resource
    resource:
      name: cpu
      target:
        type: Utilization
```



networkpolicy.yaml



resource-limit.yaml

```
ubuntu@ip-172-31-17-143:~/food-ordering-k8s/yaml$ cat resource-limit.yaml
apiVersion: v1

kind: LimitRange

metadata:

  name: cpu-memory-limit

  namespace: food-app

spec:

  limits:

  - default:

      cpu: "500m"

      memory: "512Mi"

    defaultRequest:

      cpu: "200m"

      memory: "256Mi"

    type: Container
ubuntu@ip-172-31-17-143:~/food-ordering-k8s/yaml$
```

kubectl apply -f namespace.yaml

kubectl apply -f secret.yaml

kubectl apply -f configmap.yaml

kubectl apply -f mysql-deployment.yaml

kubectl apply -f mysql-service.yaml

kubectl apply -f app-deployment.yaml

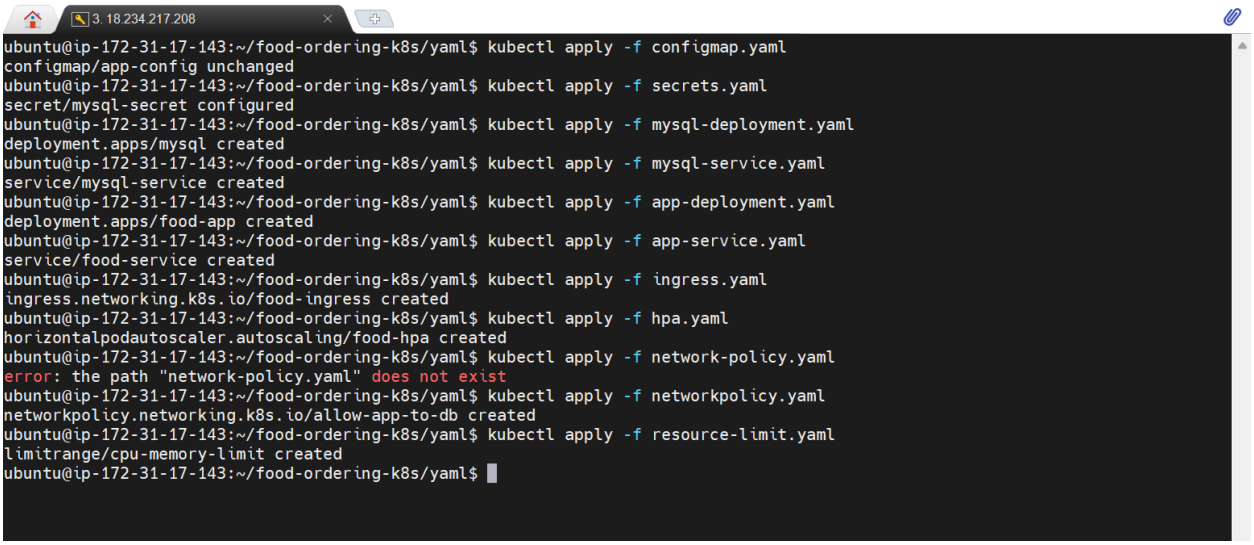
kubectl apply -f app-service.yaml

kubectl apply -f ingress.yaml

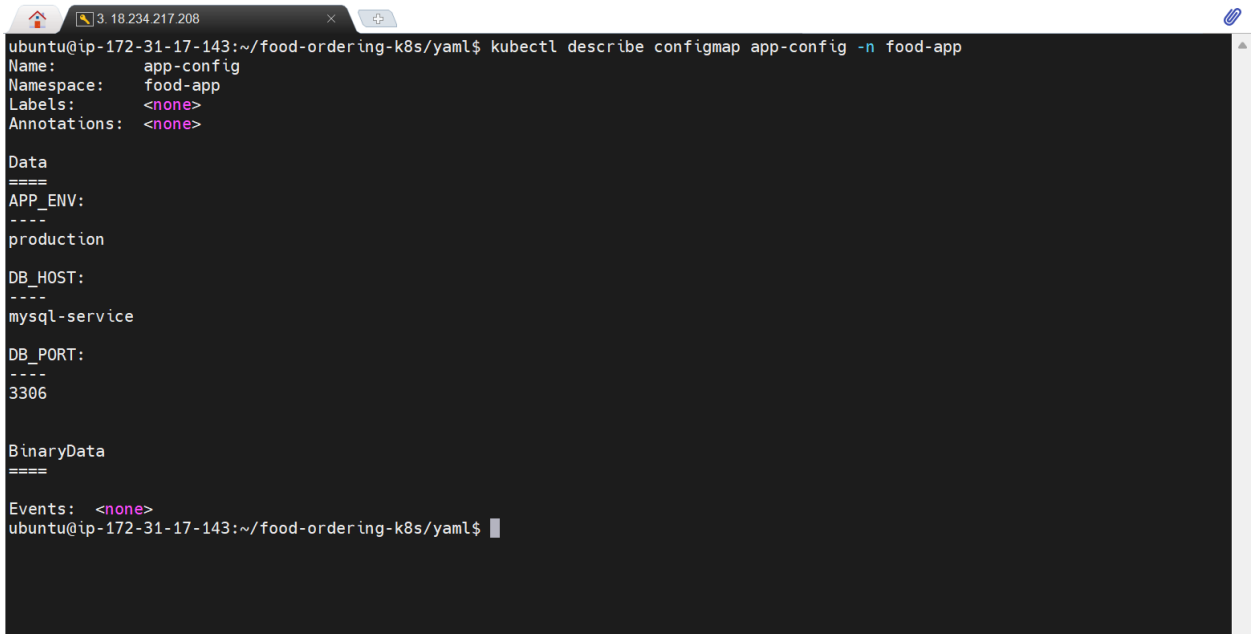
kubectl apply -f hpa.yaml

kubectl apply -f network-policy.yaml

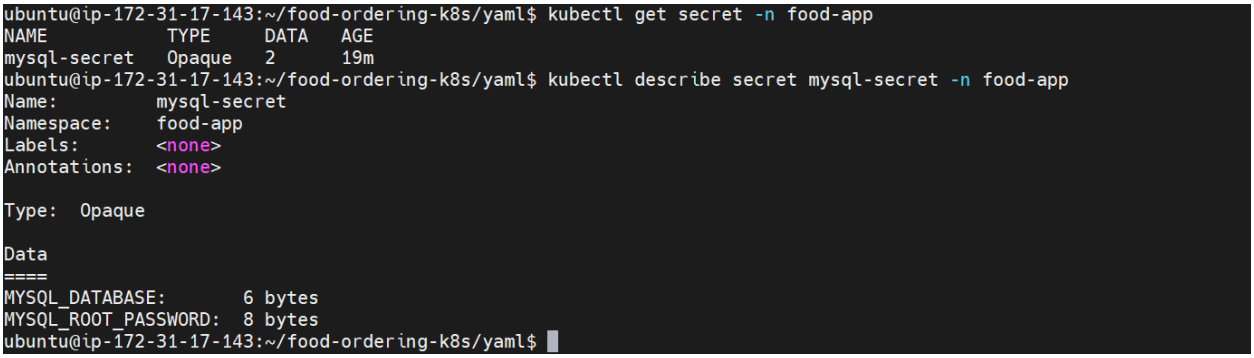
kubectl apply -f resource-limit.yaml



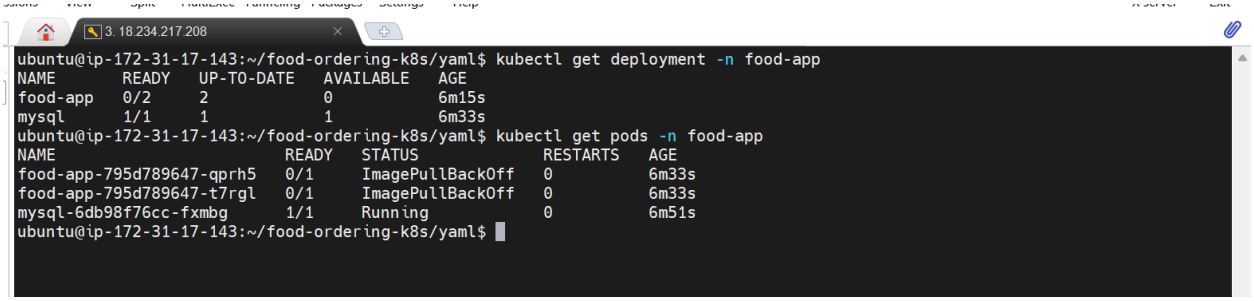
Now verify everything



Secrets



Here I got imagepullbackoff error due to the wrong name make sure you have given container name correct and image name correct



```
buntu@ip-172-31-33-155:~$ kubectl get all -n food-app
AME                                  READY   STATUS                       RESTARTS   AGE
od/cafe-website-74598fc9f9-wccfb      1/1     Running                      0          5m13s
od/cafe-website-74598fc9f9-xvh5l      1/1     Running                      0          5m13s
od/mysql-6db98f76cc-5nz28             0/1     CreateContainerConfigError   0          16m

AME                     TYPE       CLUSTER-IP       EXTERNAL-IP   PORT(S)        AGE
ervice/food-service     NodePort   10.109.148.178   <none>        80:30993/TCP   5m6s

AME                              READY   UP-TO-DATE   AVAILABLE   AGE
eployment.apps/cafe-website      2/2     2            2           5m13s
eployment.apps/mysql             0/1     1            0           16m

AME                                        DESIRED   CURRENT   READY   AGE
eplicaset.apps/cafe-website-74598fc9f9     2         2         2       5m13s
eplicaset.apps/mysql-6db98f76cc            1         1         0       16m

AME                                            REFERENCE             TARGETS              MINPODS   MAXPODS   REPLICAS   AGE
orizontalpodautoscaler.autoscaling/food-hpa    Deployment/food-app   cpu: <unknown>/50%   2         6         0          88s
buntu@ip-172-31-33-155:~$
```

```
ubuntu@ip-172-31-33-155:~$ kubectl apply -f secret.yaml
secret/mysql-secret created
ubuntu@ip-172-31-33-155:~$ kubectl rollout restart deployment mysql -n food-app
deployment.apps/mysql restarted
ubuntu@ip-172-31-33-155:~$ kubectl get pods -n food-app
NAME                              READY   STATUS        RESTARTS   AGE
cafe-website-74598fc9f9-wccfb     1/1     Running       0          10m
cafe-website-74598fc9f9-xvh5l     1/1     Running       0          10m
mysql-645b6f7cf-vgqz6             1/1     Running       0          12s
mysql-6db98f76cc-5nz28            1/1     Terminating   0          21m
ubuntu@ip-172-31-33-155:~$
```

8.54.158.57.71

```
ubuntu@ip-172-31-33-155:~$ kubectl get secrets -n food-app
No resources found in food-app namespace.
ubuntu@ip-172-31-33-155:~$ kubectl apply -f secret.yaml
secret/mysql-secret created
ubuntu@ip-172-31-33-155:~$ kubectl rollout restart deployment mysql -n food-app
deployment.apps/mysql restarted
ubuntu@ip-172-31-33-155:~$ kubectl get pods -n food-app
NAME                              READY   STATUS        RESTARTS   AGE
cafe-website-74598fc9f9-wccfb     1/1     Running       0          10m
cafe-website-74598fc9f9-xvh5l     1/1     Running       0          10m
mysql-645b6f7cf-vgqz6             1/1     Running       0          12s
mysql-6db98f76cc-5nz28            1/1     Terminating   0          21m
ubuntu@ip-172-31-33-155:~$ curl ifconfig.me
54.158.57.71ubuntu@ip-172-31-33-155:~$ ^C
ubuntu@ip-172-31-33-155:~$ kubectl get svc -n food-app
NAME           TYPE       CLUSTER-IP       EXTERNAL-IP   PORT(S)        AGE
food-service   NodePort   10.109.148.178   <none>        80:30993/TCP   11m
ubuntu@ip-172-31-33-155:~$ ^C
ubuntu@ip-172-31-33-155:~$ kubectl get endpoints -n food-app
Warning: v1 Endpoints is deprecated in v1.33+; use discovery.k8s.io/v1 EndpointSlice
NAME           ENDPOINTS   AGE
food-service   <none>      12m
ubuntu@ip-172-31-33-155:~$
```

```
buntu@ip-172-31-33-155:~$ kubectl get endpoints -n food-app
arning: v1 Endpoints is deprecated in v1.33+; use discovery.k8s.io/v1 EndpointSlice
AME           ENDPOINTS                         AGE
ood-service   10.244.0.4:80,10.244.0.5:80   15m
buntu@ip-172-31-33-155:~$ ^C
buntu@ip-172-31-33-155:~$ kubectl get svc -n food-app
AME           TYPE       CLUSTER-IP       EXTERNAL-IP   PORT(S)        AGE
ood-service   NodePort   10.109.148.178   <none>        80:30993/TCP   16m
buntu@ip-172-31-33-155:~$
```

Here the port is 3099 now add the port in the security groups

```
food-service   NodePort   10.109.148.178   <none>        80:30993/TCP   16m
ubuntu@ip-172-31-33-155:~$ ^C
ubuntu@ip-172-31-33-155:~$ curl http://localhost:30993
curl: (7) Failed to connect to localhost port 30993 after 0 ms: Could not connect to server
ubuntu@ip-172-31-33-155:~$ ^C
ubuntu@ip-172-31-33-155:~$
```

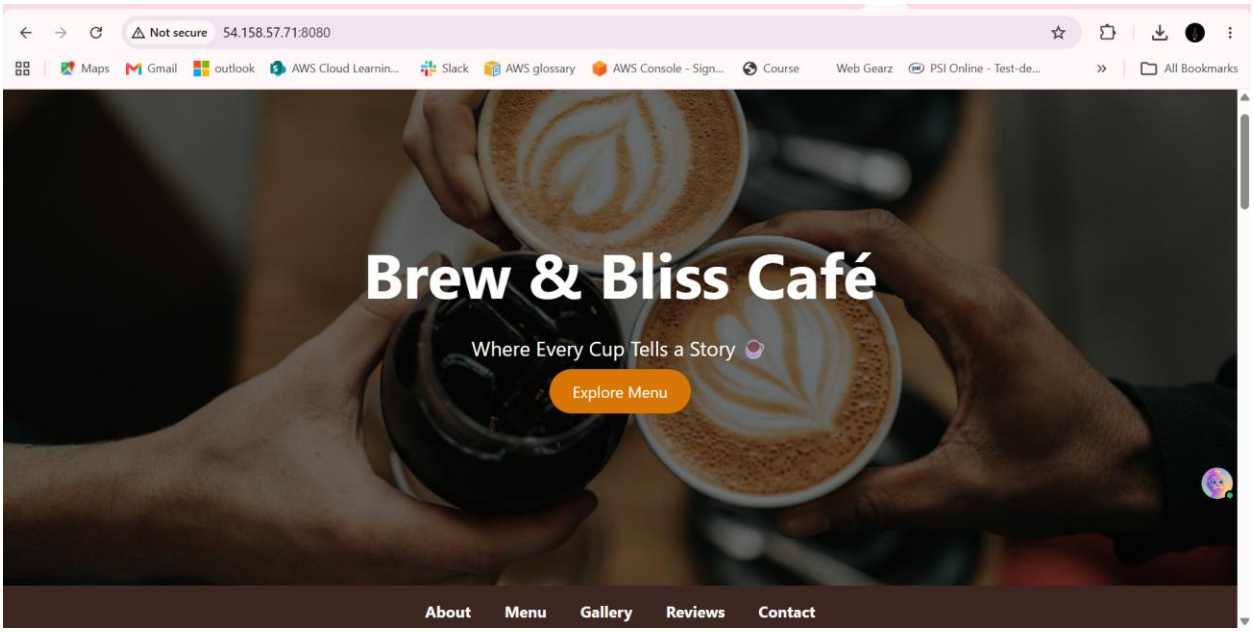
Here we can access the website

```
ubuntu@ip-172-31-33-155:~$ minikube ip
192.168.49.2
ubuntu@ip-172-31-33-155:~$ ^C
ubuntu@ip-172-31-33-155:~$ ^C
ubuntu@ip-172-31-33-155:~$ ^C
ubuntu@ip-172-31-33-155:~$ minikube service food-service -n food-app --url
http://192.168.49.2:30993
ubuntu@ip-172-31-33-155:~$ ^C
ubuntu@ip-172-31-33-155:~$ curl http://192.168.49.2:30993
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Brew & Bliss Café</title>
    <link rel="stylesheet" href="styles.css">
</head>
```

Port forwading because it cant be accessed in the web because of node port

```
ubuntu@ip-172-31-33-155:~$ kubectl port-forward -n food-app service/food-service 8080:80 --address 0.0.0.0
Forwarding from 0.0.0.0:8080 -> 80
Handling connection for 8080
^Cubuntu@ip-172-31-33-155:~$
```

Now enter your ec2 ip with http you will see the website in your web browser


Not secure
54.158.57.71:8080
Maps
Gmail
outlook
AWS Cloud Learnin...
Slack
AWS glossary
AWS Console - Sign...
Course
Web Gearz
PSI Online - Test-de...
All Bookmarks
Brew & Bliss Café
Where Every Cup Tells a Story
Explore Menu
About
Menu
Gallery
Reviews
Contact